Project Proposal

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Introduction

We will investigate a famous problem in the calculus of variations known as the *brachistochrone* ("least time") problem, originally formulated by John Bernoulli in 1696.

Problem 1. *Find the curve of fastest gravitational descent, i.e. the following minimizer:*

$$\min_{x(t)} \ \frac{1}{\sqrt{2g}} \int_{t_1}^{t_2} \sqrt{\frac{1 + x'(t)^2}{x(t)}} \ dt$$

We can solve this problem using a plethora of modern techniques that were not known at the time of its formulation; in particular, it was not known how to take the derivative of a function.

It's different from other problems studied in this course in that it's an unconstrained non-linear optimization problem with a now-known unique, efficient solution; perhaps more interesting, we are asked to find a curve instead of an n-dimensional vector.

Problem

We will discuss a historical method of solution (e.g. the methods due to Newton, or to Euler, or to Lagrange/Weierstrass) as well as a modern method (e.g. using the Karush-Kuhn-Tucker optimality conditions). We will find that the curve traces an arc of a circle, and that no other chord can perform any better.

Once we analytically derive a solution, we will computationally verify its optimality by numerically computing the integrals of other elementary functions (e.g. parabola, line, Gaussian function) This will require further investigation of the solution methods as they hint at methods of numerical integration. In particular, these methods of numerical integration can be performed even for curves with no elementary analytic antiderivative (as is the case for many formulas involving the Gaussian function).

Other Problems

After we analytically derive a solution and experimentally confirm optimality, we will attempt to derive a solution through other historical methods.

We may also explore other methods of numerical integration.